Table 1: E. coli + P. putida datasets (pre-build, 2026-10-15) built as LMDBs in the TorchCell dev tree, with a description of each. *Description* is written from the loader module and its note. *Sequence-only fit* and *Label unit* are derived from the genotype of 400 evenly spaced records per store: how well a sequence-embedding-only model fits the dataset, and what one label attaches to without the cell graph. The other columns are as in the non-expanded table.

Dataset	Host	Description	Genotypes	Env	Instances	Phenotype	Shape	Graph role	Signal (gzip, bits)	Sequence-only fit	Label unit
Environmental / chemogenomic response											
CrisprPineneToleranceNiu2019	*E. coli*	Niu 2019 tested up- and down-regulated pinene-response genes one at a time in *E. coli* BW25113(PT5-dxs) with a dCas9 activator and repressor, reading OD600 after 12 h in LB plus 0.5% pinene as a ratio against the no-guide control. This store holds the growth column, 54 records stored as a log2 ratio, one per targeted gene set and arm.	54	1	54	environment__response (log2__ratio)	scalar	global	4.0×10^{4}	best	1 gene (1-6 edits)
CrispriChemgenChoe2025	*E. coli*	Choe 2025 grew a 39,591-member sgRNA library covering all 4,198 MG1655 coding sequences in *E. coli* MG1655 carrying pdCas9, in LB and in LB plus a sub-inhibitory dose of each of twelve antibiotics, reading guide abundance by amplicon sequencing. One of the 470,502 records is one guide in one antibiotic, a log2 ratio of abundance against untreated LB.	39,291	12	470,502	environment__response (log2__ratio)	scalar	global	2.4×10^{8}	good	1 gene x condition
CrispriGeneFitnessWang2018	*E. coli*	Wang 2018 pooled a genome-scale library of 55,671 sgRNAs through five CRISPRi selections in *E. coli* MG1655: essentiality, auxotrophy, L-Trp biosynthesis, furfural tolerance and isobutanol tolerance. This store holds Supplementary Data 5, the median guide log2 fitness per BLASTN cluster with its Z score. One of the 20,158 records is one gene in one screen.	13,641	5	20,158	environment__response (log2__ratio)	scalar	global	1.5×10^{7}	good	1 gene x condition (1-7 edits)
CrispriGuideFfaEnrichmentFang2025	*E. coli*	Fang 2025 ran Wang 2018's genome-scale CRISPRi library through two rounds of Nile-Red fluorescence-activated cell sorting in an *E. coli* free-fatty-acid producing strain, counting guides by NGS before and after each sort. One of the 15,708 records is one guide in one round, a log2 enrichment of normalized reads after over before sorting.	15,711	1	15,711	environment__response (log2__ratio)	scalar	global	7.3×10^{6}	best	1 gene (1-5 edits)
CrispriGuideFitnessWang2018	*E. coli*	Wang 2018 ran a genome-scale library of 55,671 sgRNAs plus 400 non-targeting controls through five pooled CRISPRi selections in *E. coli* MG1655: essentiality in LB, auxotrophy and L-Trp biosynthesis in MOPS, and furfural and isobutanol tolerance. One of the 240,481 records is one guide in one screen, a log2 read-count ratio centered on the control median.	53,203	5	240,481	environment__response (log2__ratio)	scalar	global	1.6×10^{8}	good	1 gene x condition
CrispriKnockdownCui2018	*E. coli*	Cui 2018 electroporated a library of about 92,000 sgRNAs targeting random NGG-PAM positions of the MG1655 chromosome into two MG1655 derivatives that differ only in dCas9 level, and grew each pool for 17 generations under anhydrotetracycline induction. One record is one guide in one screened strain, the released log2 fold change in guide abundance.	71,696	1	143,392	environment__response (log2__ratio)	scalar	global	7.1×10^{7}	best	1 gene (1-7 edits)
CrispriScreenRousset2018	*E. coli*	Rousset 2018 screened a pooled library of about 92,000 random-position sgRNAs in two *E. coli* MG1655 derivatives carrying inducible dCas9, releasing per-guide log2 fold changes for five screens: growth over 17 generations, challenge by phage lambda, T4 and 186cIts at MOI 1, and cosmid recovery after lambda transduction. One of the 92,252 records is one guide in one screen.	40,592	4	92,252	environment__response (log2__ratio)	scalar	global	4.1×10^{7}	good	1 gene x condition
DoublingTimeCaglar2017	*E. coli*	Caglar 2017 Table S5 releases the exponential-phase doubling time of wild-type *E. coli* B REL606 in 19 environments, taken as log base e of 2 over the slope of a linear fit to OD600. One record is one biological replicate growth curve, 55 in all, storing the doubling time in minutes with the released asymmetric 95% confidence limits.	1	16	55	environment__response (growth__rate)	scalar	global	3.2×10^{4}	no	unperturbed strain readout x condition
EnvChemgenGirgis2009	*E. coli*	Girgis 2009 propagated a library of about 5 x 10^5 single-insertion transposon mutants of *E. coli* MG1655 for several days in each of 17 antibiotics at growth-impairing doses, then read the surviving population by microarray genetic footprinting. One of the 66,287 records is one gene in one antibiotic, carrying the combined z-score of Dataset S5.	3,976	17	66,287	environment__response (z__score)	scalar	global	3.6×10^{7}	good	1 gene x condition
EnvChemgenNichols2011	*E. coli*	Nichols 2011 pinned the Keio deletion collection, tagged and allele versions of essential genes and an sRNA library at 1,536 colonies per plate onto agar carrying one stress each, imaged the plates and scored every colony size. Table S2 releases 3,979 strain rows by 324 screen columns; one of the 1,286,962 records is one strain in one screen.	3,967	324	1,286,962	environment__response (z__score)	scalar	global	4.0×10^{8}	good	1 gene x condition
EnvChemgenShiver2016	*E. coli*	Shiver 2016 pinned an arrayed *E. coli* K-12 mutant library onto agar plates carrying one stress each, quantified colony opacity with Iris, and assigned each mutant a signed fitness-score per condition. This store serves the study's own 57 conditions of S1 Dataset; one of the 210,998 records is one mutant strain in one condition.	3,846	51	210,998	environment__response (z__score)	scalar	global	9.9×10^{7}	good	1 gene x condition
EnvChemgenWang2015	*E. coli*	Wang 2015 grew the parent BW25113 and 46 of its Keio single-gene deletions, 45 multidrug-transporter genes plus tolC, in 2YT with and without 0.5% (v/v) isoprenol at 30 C and read OD600 after 12 h. One of the 46 records is one deletion strain, stored as the log2 of its Table S3 tolerance relative to the parent.	46	1	46	environment__response (log2__ratio)	scalar	global	2.9×10^{4}	best	1 gene
EnvChemgenWang2024	*E. coli*	Wang 2024 grew a Tn5 insertion library of about 310,000 insertions in MG1655 to mid-exponential phase in LB, dosed three flasks with 2, 32 and 160 mg/L rifampicin, and recovered survivors 1 and 3 h later, comparing treated and input pools gene by gene by Tn-seq and TRANSIT resampling. One of the 26,514 records is one gene in one dose and time.	4,419	6	26,514	environment__response (log2__ratio)	scalar	global	9.3×10^{6}	good	1 gene x condition
FitnessBrowserEcoli	*E. coli*	Price 2021 released six GapMind samples through the Fitness Browser: a Keio ML9a transposon library grown on a defined no-carbon base with D-mannitol at 10, 5 and 2.5 mM as the sole carbon source, in duplicate. This loader serves the browser's BW25113 experiments that no other release covers; one record is one released locus in one sample, a signed log2 gene fitness.	3,789	3	22,734	environment__response (log2__ratio)	scalar	global	1.1×10^{7}	good	1 gene x condition
GrowthRateLamoureux2023	*E. coli*	Lamoureux 2023 growth rates from the PRECISE-1K *E. coli* RNA-seq sample metadata: wild-type MG1655, gene deletions, rpoB substitutions, evolved isolates, BW25113 and pColi construct samples across many conditions. One record is one released sample's absolute growth rate in 1/hr, 354 in all, against a wild-type M9 glucose base condition.	21	54	354	environment__response (growth__rate)	scalar	global	1.9×10^{5}	no	unperturbed strain readout x condition (0-1 edits)
GrowthRateS23Schmidt2016	*E. coli*	Schmidt 2016 Table S23 growth rates of wild-type *E. coli* BW25113, MG1655 and NCM3722 across the growth conditions of the proteome map, including chemostat and stationary-phase arms. One record is one strain in one condition, its absolute growth rate in 1/h with the released Stdev, against that strain's glucose row.	1	22	26	environment__response (growth__rate)	scalar	global	2.0×10^{4}	no	unperturbed strain readout x condition
GrowthYieldBaba2006	*E. coli*	Baba 2006 Table 3 growth yields of the obtained *E. coli* Keio single-gene deletion mutants of BW25113: OD600 after 22 h in LB with kanamycin and after 24 h and 48 h in 0.4% glucose MOPS with kanamycin. One record is one mutant in one of those three cultures, 11,735 single readings with no released replicate dispersion.	3,912	3	11,735	environment__response (optical__density)	scalar	global	3.3×10^{6}	good	1 gene x condition
MismatchCrispriFitnessHawkins2020	*E. coli*	Hawkins 2020 mismatch-CRISPRi in *E. coli*: essential genes knocked down by dCas9 with fully complementary and singly mismatched sgRNAs, competed as a pool in LB with 1 mM IPTG at 37 C for about 10 doublings. One record is one sgRNA, its mean relative fitness over 4 biological replicates, 24,186 records; the predicted guide activity is not stored.	24,186	1	24,186	environment__response (relative__growth__rate)	scalar	global	1.5×10^{7}	best	1 gene
PhageRbTnseqMutalik2020	*E. coli*	Mutalik 2020 RB-TnSeq of the *E. coli* BW25113 transposon library KEIO ML9 challenged with 14 double-stranded DNA phages at a range of multiplicities of infection, in liquid and on solid agar, read out by Bar-seq. One record is one gene in one experiment, its normalized log2 abundance change, over 68 phage assays and 10 no-phage controls.	3,716	71	287,815	environment__response (log2__ratio)	scalar	global	2.1×10^{8}	good	1 gene x condition
RbTnseqPrice2018Ecoli	*E. coli*	Price 2018's genome-wide mutant fitness compendium, *E. coli* BW25113 arm: the randomly barcoded Tn5 library KEIO ML9 assayed in the 162 successful BarSeq samples. One record is one gene in one sample, the signed log2 gene fitness with the paper's standard error, the maximum of its strain-consistency and read-count estimates.	3,789	104	598,662	environment__response (log2__ratio)	scalar	global	4.0×10^{8}	good	1 gene x condition
RbTnseqTeteneva2024	*E. coli*	Teteneva 2024 incubated a randomly barcoded Tn5 library of *E. coli* K-12 W3110 RpoS+ in lake water, filter-sterilized or not, sampling at days 2, 4 and 8. One record is one released Table S4 value, 66,162 in all: a gene's log2 barcode-count fitness against the day-0 culture in one water sample, filtration and day.	3,691	18	66,162	environment__response (log2__ratio)	scalar	global	2.3×10^{7}	good	1 gene x condition
RbTnseqWetmore2015	*E. coli*	Wetmore 2015, the RB-TnSeq method paper, grew the *E. coli* BW25113 Tn5 library KEIO ML9 in 101 condition samples, of which 92 passed the paper's quality rules. One record is one released gene in one kept sample: the normalized signed log2 gene fitness with its per-gene standard error and the released moderated t statistic.	3,646	85	353,662	environment__response (log2__ratio)	scalar	global	2.4×10^{8}	good	1 gene x condition
EnvMetalTnseqRoyet2025	*P. putida*	Royet 2025 mariner Tn-seq of KT2440: one saturated, non-barcoded transposon library grown 12 generations in LB at 28 C and in LB plus a sub-inhibitory dose of cobalt, copper, zinc or cadmium chloride. One record is one gene in one metal, TRANSIT's log2 fold change of insertion reads against LB over two pools; 21,583 of 22,916 cells are stored.	5,491	4	21,583	environment__response (log2__ratio)	scalar	global	6.7×10^{6}	good	1 gene x condition
FitnessBrowserPutida	*P. putida*	Fitness Browser RB-TnSeq of KT2440, keyed by publication: 234 samples of ten attributed studies, 1,118,052 records, one per locus and sample. One record is the browser's gene fitness for that locus's transposon insertion mutants in the sample's own barcoded library, with the moderated t statistic, under the sample's medium and carbon, nitrogen or stress variable.	9,556	146	1,118,052	environment__response (log2__ratio)	scalar	global	5.3×10^{8}	good	1 gene x condition
IsoprenolSelectionMenasalvas2025	*P. putida*	Menasalvas 2025 growth-coupled selection: a pooled dCpf1 CRISPRi library, three guides per gene, run through two sequential rounds of an isoprenol biosensor whose mCherry reporter was replaced by pyrF, on an isoprenol-producing KT2440 chassis. 58 records, one per enriched knockdown target of Supplementary Tables 1 and 2, stored as a categorical enriched call.	58	1	58	environment__response (categorical)	scalar	global	7.5×10^{4}	weak	set of 8 edits incl. heterologous genes (8-10 edits)
IsoprenolToleranceLim2025	*P. putida*	Lim 2025 isoprenol tolerance from a 16-lineage tolerization evolution campaign on KT2440 and two stacked deletion strains. Three records: IPL300 and IPL400 at the 4 g/L starting dose from Supplementary Table 3's initial growth rates, and a reverse-engineered single-deletion strain at 6 g/L, each a log2 growth-rate ratio to KT2440 by the liquid-OD assay.	3	2	3	environment__response (log2__ratio)	scalar	global	1.8×10^{4}	good	1 gene x condition (1-11 edits)
LactamGrowthRateThompson2019	*P. putida*	Thompson 2019 Supplementary Table S1: the maximal specific growth rate in per hour of KT2440 and two deletion mutants on each of three 10 mM carbon sources, glucose, 5-aminovalerate and valerolactam, grown plasmid free. Nine records, one per strain on one carbon source, including three released zeros; each reference is the same strain on glucose.	3	3	9	environment__response (growth__rate)	scalar	global	1.2×10^{4}	no	unperturbed strain readout x condition (0-2 edits)
RbTnseqBorchert2023	*P. putida*	Borchert 2023 lignin tolerance RB-TnSeq: 14 experiments in biological triplicate on a barcoded KT2440 transposon library, M9 with 20 mM glucose against eleven lignin-relevant stressors. 10,824 records over the 271 loci the Borchert 2024 compendium eliminated, one per locus and replicate arm, each a signed log2 gene fitness with its growth duration.	271	32	10,824	environment__response (log2__ratio)	scalar	global	4.5×10^{6}	good	1 gene x condition
RbTnseqBorchert2024	*P. putida*	Borchert 2024 fModules compendium: 332 RB-TnSeq samples of a barcoded KT2440 transposon library over 4,732 genes, with replicates not averaged. 1,372,280 records from the 290 kept samples, one per gene and sample, each the signed log2 gene fitness of that gene's insertion mutants under the sample's medium plus its carbon, nitrogen or stress variable.	9,464	155	1,466,920	environment__response (log2__ratio)	scalar	global	4.2×10^{8}	good	1 gene x condition
Expression (RNA-seq)											
MrnaFractionBalakrishnan2022	*E. coli*	Balakrishnan 2022 RNA-seq transcriptome of *E. coli* NCM3722 and three NCM3722 derivatives at steady state across three growth-limitation series (C-, A- and R-limitation). One record is one released sample column, 28 in all, holding the mRNA number fraction of each of 4,342 genes, a dimensionless value that sums to 1 over the sample.	3	14	28	mrna__number__fraction (rnaseq__mrna__number__fraction)	vector (4342)	node	7.2×10^{6}	best	1 gene, unperturbed strain x condition (0-2 edits)
RnaseqCaglar2017	*E. coli*	Caglar 2017 measured mRNA by RNA-seq in wild-type *E. coli* REL606 across 34 Davis minimal medium conditions: four carbon sources, Mg2+ and Na+ series, exponential and stationary phase, and two starvation time courses. One record is one Table S2 library, 152 in all, holding read counts and TPM for 4,196 protein-coding genes.	1	57	152	expression__tpm (rnaseq__tpm)	vector (4196)	node	6.7×10^{7}	best	1 gene, unperturbed strain x condition
RnaseqGuzman2018	*E. coli*	Guzman 2018 released 14 RNA-seq libraries through its own GEO submission: two wild-type BW25113 libraries and twelve of Keio thrA, serB and ptsI deletion strains that grew after extended culture on glucose M9, each in biological duplicate. One record is one library, holding the released TPM matrix keyed by the release's own gene ids.	7	2	14	expression__tpm (rnaseq__tpm)	vector (4329)	node	4.5×10^{6}	good	readout gene x 3 perturbed gene(s) x condition (0-5 edits)
RnaseqLamoureux2023	*E. coli*	PRECISE-1K (Lamoureux 2023) assembled 1,035 high-quality *E. coli* K-12 MG1655 RNA-seq profiles, all generated by one lab under one protocol and released as log2 of TPM plus 1 over 4,257 genes with per-sample metadata. One record is one sample, a biological-replicate library of a wild type or deletion strain, with TPM and read counts.	74	119	1,035	expression__tpm (rnaseq__tpm)	vector (4257)	node	5.8×10^{8}	best	1 gene, unperturbed strain x condition (0-3 edits)
RnaseqPublicK12Lamoureux2023	*E. coli*	Lamoureux 2023 also curated every publicly available *E. coli* K-12 RNA-seq run in the SRA and reprocessed it through the PRECISE-1K pipeline, keeping the 1,675 samples that passed the replicate-correlation threshold. One record is one reprocessed library of another lab, identified by its SRA experiment accession.	72	198	1,675	expression__tpm (rnaseq__tpm__from__released__counts)	vector (4355)	node	9.5×10^{8}	best	1 gene, unperturbed strain x condition (0-4 edits)
PutidaPrecise321Lim2022	*P. putida*	Lim 2022 putidaPRECISE321: 321 RNA-seq profiles of KT2440 and its derivatives over 118 conditions and 21 projects, 225 reprocessed from other studies and 96 generated for this paper. One record is one sample over 5,564 genes, carrying TPM back-transformed from the published log2(TPM + 1) matrix beside the paired featureCounts counts.	14	118	321	expression__tpm (rnaseq__tpm)	vector (5564)	node	2.4×10^{8}	best	1 gene, unperturbed strain x condition (0-2 edits)
Fitness											
CarbonSourceTong2020	*E. coli*	Tong 2020 pinned the Keio deletion collection and a small RNA and small protein deletion library at 1,536 colonies per plate on solid MOPS minimal agar carrying one carbon source, scanning every 20 min for 24 h at 37 C. Table S1A releases the normalized 24 h end-point growth of 3,796 strains on 30 carbon sources. One record is one deletion strain on one carbon source.	3,714	30	111,420	fitness	scalar	global	4.8×10^{7}	good	1 gene x condition
CrispriCrossRachwalski2024	*E. coli*	Rachwalski 2024 conjugated an arrayed collection of mobile CRISPRi plasmids targeting essential *E. coli* genes into BW25113 and BW25113 lpp, and three single knockdowns into the Keio deletion collection, reading normalized colony growth on solid LB or MOPS-glucose at 37 C. One of the 48,900 records is one strain in one medium at one anhydrotetracycline dose.	14,209	15	48,900	fitness	scalar	global	1.9×10^{7}	good	gene pair, strain readout x condition (1-2 edits)
GrowthAucRapp2026	*E. coli*	Rapp 2026 growth of an arrayed *E. coli* CRISPRi library (dCas9 host YYdCas9) covering the 1,515 genes of the iML1515 model: the area under each strain's OD600 curve over three replicate cultures, divided by the mean AUC of 48 control curves. One record is one knockdown strain, 1,515 in all, 18 of them without a metabolome sample.	1,515	1	1,515	fitness	scalar	global	7.7×10^{5}	best	1 gene
GrowthRateCampos2018	*E. coli*	Campos 2018 growth of 3,664 imaged strains of the *E. coli* Keio single-gene deletion collection, grown in 96-well plates in M9 with 0.1% casamino acids and 0.2% glucose at 30 C. One record is one deletion strain, its Gompertz maximal growth rate as a ratio to the BW25113 parent, of which 240 replicates were grown alongside.	3,664	1	3,664	fitness	scalar	global	1.1×10^{6}	best	1 gene
GrowthRateChoe2019	*E. coli*	Choe 2019 growth rates from the genome-reduced *E. coli* MS56 study, Source Data Fig. 2c and 2f: the MS56 Δ21kb and ΔrpoS deletions, three single-allele strains (cspC, ilvN, yifB) against kan-marked controls, MS56 over MG1655 and the evolved clone eMS57 over MS56. One record is one strain as a ratio to its parent, eight in all.	6	1	8	fitness	scalar	global	1.1×10^{4}	best	1 gene (0-21 edits)
GrowthRateSchmidt2016	*E. coli*	Schmidt 2016 Table S24 growth rates of the Keio rimI, rimJ and rimL deletion strains of *E. coli* BW25113 in glucose and in acetate medium, with up to three replicates each. One record is one deletion strain in one medium, its growth rate as a ratio to the wild type in the same medium, with the sample SD; six records in all.	3	2	6	fitness	scalar	global	7.8×10^{3}	good	1 gene x condition
GrowthWang2015	*E. coli*	Wang 2015 Table S3 growth of 46 *E. coli* Keio transporter deletion strains of BW25113 in plain 2YT medium without isoprenol, read as 12 h endpoint OD600. One record is one deletion strain, its OD600 as a ratio to the BW25113 parent with the scaled sample SD; the isoprenol tolerance arm is a separate dataset.	46	1	46	fitness	scalar	global	2.1×10^{4}	best	1 gene
TranscriptionFactorKnockoutChoe2019	*E. coli*	Choe 2019 monitored growth of 62 transcription-factor knockout strains of the Keio collection in M9 glucose medium, in a 96-well plate on a microplate reader at 37 C. Only two carry a released number, so one record is one of those deletions: ΔydaS and ΔabgR, at 72.4% and 69.8% of wild-type BW25113.	2	1	2	fitness	scalar	global	5.3×10^{3}	best	1 gene
FitnessContrastBorchert2023	*P. putida*	Borchert 2023 two-arm RB-TnSeq fitness contrasts of the barcoded KT2440 transposon library, released as 13 comparison sheets with three replicate cultures per arm. 64,853 records, one per gene in one comparison of a condition arm against a reference arm, carrying the released t statistic, p value, q value and adjusted q value of the two-sample t test.	5,031	12	64,853	t__statistic (rbtnseq__gene__fitness__two__sample__t__test)	scalar	global	2.2×10^{7}	good	1 gene x condition
Flux											
FluxIshii2007	*E. coli*	Ishii 2007 ran 13C metabolic flux analysis on one set of glucose-limited chemostat cultures of *E. coli* K-12: 24 Keio single-gene disruptants at 0.2 per hour plus the wild type at 0.1, 0.4, 0.5 and 0.7 per hour. One of the 28 records is one culture's fitted net-flux map over the release's own 43 net reactions, as a percentage of the glucose uptake rate.	25	5	28	net__flux (c13__mfa__net__flux__percent__of__specific__glucose__uptake)	vector (42)	bipartite node	6.0×10^{4}	good	reaction x 1 perturbed gene(s) x condition (0-1 edits)
FluxLi2021	*E. coli*	Li 2021 replaced the native zwf promoter of the mevalonate-producing EP-bifido strain BW-P with synthetic promoters and ran 13C metabolic flux analysis on three of them. One of the three records is one strain's released network block: 99 net fluxes in mmol/gdw/h with their 90% bounds, against the unreleased BW-P BF control map.	3	1	3	net__flux (c13__mfa__net__flux__mmol__per__gdw__per__h)	vector (99)	bipartite node	5.7×10^{4}	good	reaction x 1 perturbed gene(s)
Gene interaction											
GeneInteractionBabu2014	*E. coli*	Babu 2014 eSGA genetic interactions in *E. coli* K-12: 163 Hfr Cavalli donor query mutants were conjugated into an arrayed F- Keio recipient collection and double-mutant colony sizes were scored as a signed interaction score S. One record is one Table S2 donor-recipient gene pair, of 42,705 high-confidence pairs (25,239 aggravating, 17,466 alleviating).	42,703	1	42,705	gene__interaction	scalar	edge	1.2×10^{7}	best	gene pair
GeneInteractionButland2008	*E. coli*	Butland 2008 eSGA genetic interactions in *E. coli* K-12: 39 Hfr Cavalli query deletions were each conjugated into an arrayed F- recipient collection of 8,073 strains, Keio deletions plus SPA-tagged essential genes. One record is one cell of the unfiltered 39 x 8,073 matrix of S scores in Supplementary Table 4, so non-significant pairs are kept.	314,769	1	314,769	gene__interaction	scalar	edge	8.2×10^{7}	best	gene pair
Metabolite											
ExometaboliteBrunk2016	*E. coli*	Brunk 2016 measured glucose and five organic acids by HPLC across a batch fermentation of nine mevalonate-pathway *E. coli* DH1 strains sampled from 0 to 72 hours in EZ-Rich defined medium. This store holds the six workbook columns labeled g/L; one of the 126 records is one strain at one sampling hour, referenced to the wild-type DH1 sample of that hour.	9	28	126	metabolite__level (hplc__extracellular__concentration__g__per__l)	vector (6)	bipartite node	1.8×10^{5}	good	metabolite x 9 perturbed gene(s) x condition (0-10 edits)
MetaboliteIntensityRapp2026	*E. coli*	Rapp 2026 absolute flow-injection MS intensities from the arrayed *E. coli* CRISPRi library covering the iML1515 genes: the Table S5 annotated m/z features that accumulate in a knockdown strain, as the mean of two plate replicates with their standard error. One record is one strain, 407 records holding 1,375 values, against its batch median intensity.	407	1	407	metabolite__level (fi__ms__annotated__feature__absolute__intensity__mean__of__2__plates)	vector (2)	bipartite node	3.9×10^{5}	good	metabolite x 1 perturbed gene(s)
MetabolomeBrunk2016	*E. coli*	Brunk 2016 LC-MS metabolite concentrations from a 0 to 72 h flask time course of nine *E. coli* DH1 strains in EZ-Rich medium with 1% glucose: isopentenol, limonene and bisabolene producers carrying mevalonate-pathway plasmids, plus wild type. One record is one strain at one sampling hour, its 51 metabolites in uM, 117 records in all.	9	26	117	metabolite__level (lc__ms__concentration__um)	vector (51)	bipartite node	3.5×10^{5}	good	metabolite x 9 perturbed gene(s) x condition (0-10 edits)
MetabolomeFuhrer2017	*E. coli*	Fuhrer 2017 metabolome of the *E. coli* Keio single-gene deletion collection (BW25113), grown on glucose minimal medium with casein hydrolysate at 37 C, harvested mid-exponentially and measured by flow-injection TOF-MS over 3,169 negative-mode and 4,365 positive-mode ions. One record is one deletion strain column, 3,806 in all, with a modified z-score per ion.	3,806	1	3,806	metabolite__level (fia__tof__ms__ion__modified__z__score)	vector (7534)	bipartite node	1.9×10^{9}	good	metabolite x 1 perturbed gene(s)
MetabolomeIshii2007	*E. coli*	Ishii 2007 CE-TOFMS intracellular metabolite concentrations in mM from glucose-limited chemostat cultures of *E. coli* K-12: 24 Keio single-gene disruptants at a dilution rate of 0.2 h-1 plus the wild type at 0.1, 0.4, 0.5 and 0.7 h-1. One record is one culture, 28 in all; the proteome and 13C fluxes of the same cultures are separate datasets.	25	5	28	metabolite__level (ce__tofms__intracellular__concentration__mm)	vector (130)	bipartite node	4.9×10^{5}	good	metabolite x 1 perturbed gene(s) x condition (0-1 edits)
MetabolomeRapp2026	*E. coli*	Rapp 2026 flow-injection MS metabolome of an arrayed *E. coli* CRISPRi library covering all 1,515 iML1515 genes, with dCas9 induced by anhydrotetracycline and growth on M9 glucose at 37 C, over 3,026 extracts. One record is one knockdown strain, holding each annotated m/z feature's linear fold change over the batch median, averaged over two plates.	1,497	1	1,497	metabolite__level (fi__ms__iml1515__feature__fold__change__vs__batch__median__mean__of__2__plates)	vector (1321)	bipartite node	5.6×10^{8}	good	metabolite x 1 perturbed gene(s)
MetabolomeSchastnaya2021	*E. coli*	Schastnaya 2021 flow-injection TOF-MS metabolome of *E. coli* strains carrying MAGE phosphosite substitutions on 23 enzymes, plus matching Keio deletion strains, each grown on M9 with one of five carbon sources. One record is one strain on one carbon source: the log2 fold change of each of 464 annotated ions over the MG1655 wild type, 200 records in all.	103	5	200	metabolite__level (fia__tof__ms__ion__log2__fold__change__vs__wild__type)	vector (284)	bipartite node	3.6×10^{6}	good	metabolite x 1 perturbed gene(s) x condition
TargetedMetabolomeRapp2026	*E. coli*	Rapp 2026 screened an *E. coli* CRISPRi library, each strain the YYdCas9 background carrying one released spacer, and re-measured the accumulating strain-metabolite pairs on a second platform, the targeted LC-MS/MS screen of Table S6. One record is one strain, holding each measured metabolite's EIC peak-height fold change against the library median.	407	1	407	metabolite__level (lc__msms__eic__peak__height__fold__change__vs__library__median)	vector (2)	bipartite node	3.7×10^{5}	good	metabolite x 1 perturbed gene(s)
MetaboliteGrowthPhaseMenasalvas2025	*P. putida*	Menasalvas 2025 data S1-1, growth-phase arm: absolute intracellular metabolite concentrations in uM, averaged over three biological replicates, of the control TEAM-2595 and the two improved isoprenol producers TEAM-3174 and TEAM-3185 harvested at OD600 0.7. Three records, one per strain, each over 46 of the sheet's 48 absolute rows.	3	1	3	metabolite__level (lc__ms__intracellular__concentration__uM)	vector (46)	bipartite node	7.4×10^{4}	good	metabolite x 14 perturbed gene(s) (14-81 edits)
MetaboliteProductionPhaseMenasalvas2025	*P. putida*	Menasalvas 2025 data S1-1, production-phase arm: the same absolute uM concentrations over three biological replicates, on the control TEAM-2595 and the improved isoprenol producers TEAM-3174 and TEAM-3185 harvested at 24 h. Three records, one per strain, over 46, 46 and 47 metabolites; the two phases are two environments, hence two datasets.	3	1	3	metabolite__level (lc__ms__intracellular__concentration__uM)	vector (46)	bipartite node	7.3×10^{4}	good	metabolite x 14 perturbed gene(s) (14-81 edits)
Morphology											
MorphologyCampos2018	*E. coli*	Campos 2018 cell morphology of 3,664 imaged strains of the *E. coli* Keio single-gene deletion collection, grown in M9 with casamino acids and glucose at 30 C. One record is one deletion strain holding 26 features from Dataset EV2 (19 morphological, 2 growth, 5 cell cycle), with the number of segmented cells as its sample count.	3,664	1	3,664	morphology	vector (15)	global	8.5×10^{6}	best	1 gene
Phosphoproteome											
PhosphositeIntensityPotel2018	*E. coli*	Potel 2018 measured the *E. coli* W3110 phosphoproteome by mass spectrometry, releasing MaxQuant precursor intensities for 2,129 phosphosites (S, T, Y and H) in 12 samples: glucose or glycerol, exponential or stationary phase, three replicates. One record is one sample column carrying all 2,129 sites as released; a released 0 is the typed status not quantified.	1	2	12	phosphosite__intensity (fe3__imac__lc__msms__maxquant__summed__xic)	vector (989)	node	1.3×10^{7}	best	1 gene, unperturbed strain x condition
Product titer											
BiofuelTiterBrunk2016	*E. coli*	Brunk 2016 measured isopentenol, limonene and bisabolene titers in g/L over a batch fermentation of nine mevalonate-pathway *E. coli* DH1 strains sampled from 0 to 72 h in EZ-Rich defined medium with 1% glucose. Each strain carries one or two production plasmids whose gene parts are the perturbations. One of the 72 records is one strain at one sampling hour for its own fuel.	8	18	72	titer	scalar	global	1.1×10^{5}	weak	set of 9 edits incl. heterologous genes (7-10 edits)
IsopentenolTiterFoo2014	*E. coli*	Foo 2014 isopentenol (isoprenol) titers of *E. coli* DH1 production strains carrying the mevalonate-pathway plasmids pJBEI-6830 and pJBEI-6833 plus one of eight genes whose overexpression shortened the isopentenol growth lag, or an RFP control, induced with 500 uM IPTG. One record is one strain's 48 h titer over triplicates, nine in all.	9	1	9	titer	scalar	global	1.5×10^{4}	best	1 heterologous construct
PineneFoldChangeNiu2019	*E. coli*	Niu 2019 targeted pinene-response genes one at a time in *E. coli* BW25113(PT5-dxs) with a dCas9 MCPSoxS CRISPR activator or repressor, and quantified pinene by GC-FID after a 72 h SBMSN shake-flask run under a dodecane overlay. One record is one target gene set and arm: the released titer ratio over the no-guide control, with its sample SD over three replicates.	40	1	40	titer	scalar	global	2.2×10^{4}	best	1 gene (1-6 edits)
IsoprenolTiterBanerjee2024	*P. putida*	Banerjee 2024 isoprenol titers of a genome-scale-model-guided KT2440 campaign, taken from the article body because no released file is a measurement table. 20 records, one per strain in one condition, over five families: EZ rich plus glucose tubes, M9 minimal, a 2 L fed-batch bioreactor and sorghum hydrolysate.	14	9	20	titer	scalar	global	3.4×10^{4}	weak	set of 12 edits incl. heterologous genes (5-12 edits)
IsoprenolTiterCarruthers2025	*P. putida*	Carruthers 2025 isoprenol titer over six automated design-build-test-learn cycles of multiplexed CRISPRi arrays on the KT2440 chassis IY1449b carrying the pIY670 mevalonate pathway. 502 records: 465 campaign strain-cycles plus 37 Source Data panel strains, each with the sample SD over its triplicate cultures. The 90 control cultures are the per-cycle reference.	432	2	502	titer	scalar	global	4.2×10^{5}	weak	set of 8 edits incl. heterologous genes (6-12 edits)
IsoprenolTiterDeSiqueira2025	*P. putida*	de Siqueira 2025 Table S2, the maximum isoprenol titer in mM taken over the 24, 48 and 72 h samples of the isoprenol-catabolism-deficient strain PT and its acetate-tolerized Sigma isolates carrying the plasmid pIY670. 10 records, one per strain in one medium the library holds, each over three biological replicates with no released uncertainty.	5	2	10	titer	scalar	global	2.1×10^{5}	weak	set of 33 edits incl. heterologous genes (33-48 edits)
IsoprenolTiterYunus2026	*P. putida*	Yunus 2026 per-strain isoprenol titer from the hand-deposited Benchling table, the only released per-strain titer of this CRISPRi knockdown panel in the engineered producer IY1452, cultured 48 h in 5 mL M9 with glucose. 125 records, one per deposited strain row; the Control row's released 845.73 mg/L is the reference.	125	1	125	titer	scalar	global	6.6×10^{4}	best	1 gene
IsoprenylAcetateTiterKang2026	*P. putida*	Kang 2026 isoprenyl acetate titers by GC-FID from the isoprenol chassis PIPA engineered with a heterologous alcohol acetyltransferase, three esterase deletions, an integrated xylose isomerase pathway and an acetyl-CoA module. 26 records: seven Table 1 flask maxima, five Table S4 tube endpoints at 48 h, and the fed-batch course's ester sums and isoprenol values.	11	13	26	titer	scalar	global	6.7×10^{4}	weak	set of 20 edits incl. heterologous genes (6-20 edits)
ValerolactamTiterThompson2019	*P. putida*	Thompson 2019 valerolactam titers of a four-strain ladder, each strain a lysine-fed producer carrying the plasmid pBADT-davBA-ORF26 with lactam hydrolase, aminotransferase and racemase deletions stacked. Eight records, the four strains at 24 and 48 h; the undetectable wild-type 48 h cell is a left-censored titer at the assay floor of 0.1549 mg/L.	4	2	8	titer	scalar	global	1.7×10^{4}	good	gene triple, strain readout x condition (3-7 edits)
Promoter activity											
PromoterReporterMohiuddin2022	*E. coli*	Mohiuddin 2022 grew a 1,930-well *E. coli* MG1655 promoter-GFP reporter collection in 96-well plates, dosed early stationary phase with ampicillin, ofloxacin or gentamicin or left it untreated, and read GFP hourly from hour two to hour ten. One record is one arm, plate, well and read hour, the released plate-reader GFP value for that promoter.	1,809	27	69,480	promoter__activity	scalar	node	2.8×10^{7}	best	inserted DNA (promoter or reporter) x condition
Protein abundance											
ProteinFoldChangeCaglar2017	*E. coli*	Caglar 2017 differential proteomics from LC-MS/MS of wild-type *E. coli* REL606 grown on Davis minimal medium across carbon sources and Mg2+ and Na+ levels. One record is one Table S8 protein contrast under the paper's primary design: a growth phase and single-dose test level against the glucose base group, 8 records kept as released.	1	8	8	protein__fold__change (deseq2__wald__log2__fold__change__design__batch__plus__condition)	vector (4180)	node	1.1×10^{7}	best	1 gene, unperturbed strain x condition
ProteomeBrunk2016	*E. coli*	Brunk 2016 sampled one batch fermentation of nine *E. coli* DH1 strains, the wild type plus three isopentenol, three limonene and two bisabolene producers, from 0 to 72 hours. This class holds the selected-reaction-monitoring peak areas of all 68 host proteins, one record per strain and sampling hour, 81 records.	9	18	81	protein__abundance (srm__protein__peak__area)	vector (68)	node	3.6×10^{5}	good	readout gene x 9 perturbed gene(s) x condition (0-10 edits)
ProteomeCaglar2017	*E. coli*	Caglar 2017 measured the proteome of wild-type *E. coli* REL606 by LC-MS/MS across 34 Davis minimal medium conditions: four carbon sources, Mg2+ and Na+ series, exponential and stationary phase, and two starvation time courses. One record is one Table S3 sample, 105 in all, holding size-factor-normalized spectral counts keyed by ECB locus tag.	1	40	105	protein__abundance (lcmsms__spectral__count__deseq2__size__factor__normalized)	vector (4196)	node	2.5×10^{7}	best	1 gene, unperturbed strain x condition
ProteomeIshii2007	*E. coli*	Ishii 2007 quantified absolute protein abundances by LC-MS/MS, in mg protein per g dry cell weight, on one set of glucose-limited chemostat cultures of *E. coli* K-12. One record is one culture: 24 single-gene Keio disruptants at a dilution rate of 0.2 per hour, plus the wild type at 0.1, 0.4, 0.5 and 0.7 per hour, 28 records.	25	5	28	protein__abundance (lc__ms__ms__absolute__mg__protein__per__g__dry__cell__weight)	vector (57)	node	2.9×10^{5}	best	1 gene, its own label x condition (0-1 edits)
ProteomeMori2021	*E. coli*	Mori 2021 measured absolute *E. coli* proteome abundances over 66 LC-MS samples by DIA/SWATH with xTop protein inference, spanning nutrient limitations, non-metabolic stresses and non-planktonic states. This class stores the Dataset EV9 absolute protein mass fractions, one record per loaded sample, seven of the 66, each one MG1655 culture in one medium.	1	1	7	protein__abundance (absolute__protein__mass__fraction__xtop__dia__swath__riboprofiling__scaled)	vector (1897)	node	1.1×10^{6}	best	1 gene, unperturbed strain
ProteomeSchmidt2016	*E. coli*	Schmidt 2016 measured absolute abundances of more than 2,300 proteins of *E. coli* K-12 BW25113 across 22 growth conditions in biological triplicates by label-free LC-MS. This class stores Table S6's protein copies per cell, one record per growth condition other than the glucose reference, 21 records of wild-type cultures.	1	21	21	protein__abundance (absolute__protein__copies__per__cell__label__free__sid__anchored)	vector (2359)	node	1.1×10^{7}	best	1 gene, unperturbed strain x condition
ProteomeSrmSet1Schmidt2016	*E. coli*	Schmidt 2016 also quantified 41 carbohydrate-metabolism proteins absolutely by selected reaction monitoring against synthetic heavy reference peptides. This class holds Table S2, the OGE-fractionated arm, 41 proteins in 19 conditions with a technical-replicate standard deviation. One record is one BW25113 growth condition, 18 records.	1	18	18	protein__abundance (absolute__protein__copies__per__cell__srm__sid__dataset1__technical__sd)	vector (41)	node	1.5×10^{5}	best	1 gene, unperturbed strain x condition
ProteomeSrmSet2Schmidt2016	*E. coli*	Schmidt 2016's second absolute-quantification arm, selected reaction monitoring against synthetic heavy reference peptides on the unfractionated data set. This class holds Table S3, 31 proteins in 22 conditions with a biological-triplicate standard deviation. One record is one BW25113 growth condition, 21 records.	1	21	21	protein__abundance (absolute__protein__copies__per__cell__srm__sid__dataset2__biological__sd)	vector (31)	node	1.3×10^{5}	best	1 gene, unperturbed strain x condition
CampaignProteomeCarruthers2025	*P. putida*	Carruthers 2025 campaign proteome, read from the Dryad deposit of six automated design-build-test-learn cycles of multiplexed CRISPRi arrays on an isoprenol-producing KT2440 chassis. 465 records share the (construct, cycle) identity of the 465 titer records: one record is one strain-cycle's mean percent-of-proteome Top3 abundance, keyed by UniProt accession.	401	1	465	protein__abundance (dia__nn__top3__percent__of__proteome__mean)	vector (1616)	node	1.3×10^{8}	good	readout gene x 8 perturbed gene(s) (6-9 edits)
CrispriArrayYunus2026	*P. putida*	Yunus 2026 sgRNA-array position study, Supplementary Tables S8 to S12: 25 records, one per multiplexed CRISPRi construct built with VAMMPIRE in the engineered isoprenol producer IY1452. Each of the five panels measures one representative protein as a DIA-NN Top3 ratio to the control strain, over three biological replicates with the sample SD.	25	1	25	protein__fold__change (dia__nn__top3__relative__to__control__strain)	scalar	node	2.5×10^{4}	good	readout gene x 4 perturbed gene(s) (1-5 edits)
CrispriDifferentialProteomeYunus2026	*P. putida*	Yunus 2026 global differential proteome of the CRISPRi strain knocking down the 2-oxoglutarate dehydrogenase subunit sucB, Supplementary Tables S4 and S5, which release 145 downregulated and 193 upregulated proteins. ONE record holds the 305 of 338 released keys that resolve to a locus, each a Top3 fold change over three replicates with its t-test p value.	1	1	1	protein__fold__change (dia__nn__top3__fold__change__relative__to__control__strain__equal__variance__t__test)	vector (305)	node	6.1×10^{4}	best	1 gene, its own label
CrispriKnockdownYunus2026	*P. putida*	Yunus 2026 CRISPRi knockdown screen, Supplementary Table S3, the 125-sample shotgun-proteomics panel of FluxRETAP-chosen and intuition-chosen targets in the engineered isoprenol producer IY1452, cultured 48 h in M9 with glucose. 102 records, one per CRISPRi strain: its own target protein as a DIA-NN Top3 ratio to the control strain, from one sample.	102	1	102	protein__fold__change (dia__nn__top3__relative__to__control__strain)	scalar	node	4.9×10^{4}	best	1 gene, its own label
CrispriPanelProteomeYunus2026	*P. putida*	Yunus 2026 per-strain Top3 protein panel from the hand-deposited Benchling table: 125 records, one per deposited CRISPRi strain row of the isoprenol producer IY1452, carrying 253 UniProt accession columns beside that row's titer. One record is one strain's absolute profile over 207 accessions, with the Control row's measured proteome as the reference.	125	1	125	protein__abundance (dia__nn__top3__signal__benchling__displayed)	vector (207)	node	1.5×10^{6}	best	1 gene, its own label
ProteomeBanerjee2025	*P. putida*	Banerjee 2025 DIA shotgun proteome of the p-coumarate growth-coupling cutset in a KT2440 chassis, read from four released t-test workbooks naming six distinct groups. Six records, one per strain and medium sample: two promoter-titration strains of the rate-limiting fourth cutset gene against their parent, and its deletion strain in p-CA with alanine and malate.	4	3	6	protein__abundance (diann__lfq__log2__intensity__replicate__mean)	vector (2248)	node	2.8×10^{6}	best	1 gene, its own label x condition (0-1 edits)
ProteomeCarruthers2025	*P. putida*	Carruthers 2025 off-target study proteome, the Source Data's only abundance matrix: 20 sample groups, three replicates each, over 1,501 protein groups from IY1452b knockout strains carrying sgRNAs against candidate off-target genes. 21 records, one per released sample, each an absolute DIA-NN Top3 profile over 1,424 keys that resolve to the pinned assembly.	17	2	21	protein__abundance (dia__nn__top3__peptide__signal__mean)	vector (1424)	node	5.6×10^{6}	good	readout gene x 7 perturbed gene(s) x condition (7-9 edits)
ProteomeDeSiqueira2025	*P. putida*	de Siqueira 2025 Data Set S1 on the Top3 scale: DIA-NN Top3 peptide signal of KT2440, the isoprenol-catabolism-deficient PT and four acetate-tolerized evolved isolates, harvested in exponential growth in M9 with acetate, glucose or both. 17 records, one per sample over 1,730 protein groups with the SD over three replicates.	6	3	17	protein__abundance (dia__nn__top3__peptide__signal__replicate__mean)	vector (1531)	node	5.5×10^{6}	best	1 gene, unperturbed strain x condition (0-43 edits)
ProteomeFoldChangeCarruthers2025	*P. putida*	Carruthers 2025 per-protein log2 fold changes of the CRISPRi campaign's differential contrasts: 16 records, one per released contrast column. Figure 5b gives 14 single-guide CRISPRi strains against the non-target control strain, and Figure 6b the 2 knockout-plus-target-sgRNA strains against the same knockout with a non-targeting sgRNA.	16	1	16	protein__fold__change (dia__log2__fold__change__paired__two__tailed__t__test)	vector (350)	node	1.5×10^{6}	good	readout gene x 6 perturbed gene(s) (6-7 edits)
ProteomeFoldChangeLim2025	*P. putida*	Lim 2025 released proteome contrasts of the tolerization evolution campaign: four records, one per sheet, two evolved isolates over their parent IPL400 in M9 with 4 g/L glucose, with and without 4 g/L isoprenol. One record carries that sheet's per-protein log2 fold change with the p value, the BH adjusted p value and the Welch standard error.	2	2	4	protein__fold__change (dia__nn__top3__log2__fold__change__welch__two__sample__t__test)	vector (2361)	node	3.7×10^{6}	good	readout gene x 19 perturbed gene(s) x condition (19-21 edits)
ProteomeLim2025	*P. putida*	Lim 2025 absolute proteome under isoprenol stress: the stacked-deletion parent IPL400 and the two evolved isolates the release quantifies, profiled by shotgun proteomics under 4 g/L isoprenol. Three records, one per strain, each its Top3 log2 abundances over 2,361, 2,361 and 2,332 protein keys, referenced to its own unstressed arm.	3	1	3	protein__abundance (dia__nn__top3__log2__mean)	vector (2361)	node	1.5×10^{6}	good	readout gene x 7 perturbed gene(s) (7-21 edits)
ProteomeLog10PercentDeSiqueira2025	*P. putida*	de Siqueira 2025 Data Set S1 on its third released scale: the mean of the three replicates' log10 percent-of-total abundance with its SD, for KT2440, the acetate-sensitive PT and four tolerized isolates in M9 with acetate, glucose or both. 17 records, one per sample over 1,730 protein groups; a mean of logs cannot be recovered from the other scales.	6	3	17	protein__abundance (dia__nn__top3__log10__percent__of__total__abundance__replicate__mean)	vector (1531)	node	6.0×10^{6}	best	1 gene, unperturbed strain x condition (0-43 edits)
ProteomeMenasalvas2025	*P. putida*	Menasalvas 2025 Supplementary Data 2 sheet 4: the shotgun proteome of the control TEAM-2595 and the two improved isoprenol producers TEAM-3174 and TEAM-3185, each sampled in the growth phase at OD600 0.7 and in the production phase at 24 h. Six records, one per sample over 2,090 loci and three replicates, referenced to the matching-phase control.	3	2	6	protein__abundance (dia__protein__counts__sum__mean)	vector (2090)	node	2.6×10^{6}	good	readout gene x 14 perturbed gene(s) x condition (14-81 edits)
ProteomePercentDeSiqueira2025	*P. putida*	de Siqueira 2025 Data Set S1 as percent of each sample's total abundance, the scale the paper's own dimensionality reduction runs on, for KT2440, the acetate-sensitive PT and four tolerized isolates in M9 with acetate, glucose or both. 17 records, one per released sample over 1,730 protein groups, each a replicate-wise mean of percentages with its SD.	6	3	17	protein__abundance (dia__nn__top3__percent__of__total__abundance__replicate__mean)	vector (1531)	node	6.0×10^{6}	best	1 gene, unperturbed strain x condition (0-43 edits)
Protein synthesis / turnover											
ProteinSynthesisRateLi2014	*E. coli*	Li 2014 used ribosome profiling to quantify absolute protein synthesis rates in *E. coli* MG1655 growing in MOPS glucose medium with the full Neidhardt supplement, the same supplement without L-methionine, and no supplement. One record is one medium, wild type, holding Table S1's plain integer rates in molecules per generation for 3,041, 3,362 and 2,241 genes.	1	3	3	synthesis__rate (ribosome__profiling__footprint__density__absolute__synthesis__rate)	vector (3041)	node	3.7×10^{5}	best	1 gene, unperturbed strain x condition
ProteinTurnoverGupta2024	*E. coli*	Gupta 2024 switched a steady-state *E. coli* NCM3722 culture to 15N ammonium and followed the peak shift by TMTproC complement-reporter proteomics, fitting one turnover parameter per protein in each of 13 growth conditions. One record is one condition, holding the released total half-life in hours, for 3,262 proteins across the release.	6	8	13	degradation__rate (n15__ammonium__tmtproc__total__turnover__rate__per__hour)	vector (2553)	node	1.1×10^{7}	best	1 gene, unperturbed strain x condition (0-3 edits)
Viability											
GeneEssentialityBaba2006	*E. coli*	Baba 2006 targeted 4,288 ORFs of *E. coli* K-12 BW25113 for in-frame replacement by a kanamycin cassette and kept the mutants whose kanamycin-resistant colonies had the expected PCR structure, the Keio collection. One record is one targeted ORF on the selection plates: 3,985 yielded a deletion and 303 did not, the paper's essential gene candidates.	4,288	1	4,288	is__essential	scalar	node	1.8×10^{6}	best	1 gene, its own label
GeneEssentialityGerdes2003	*E. coli*	Gerdes 2003 mutagenized *E. coli* MG1655 with the EZ::TN KAN-2 transposome, grew the pool for 23 population doublings in a fermentor, and mapped the surviving insertions by nested PCR. Table S1 calls each ORF essential, non-essential, not covered or ambiguous; one of the 3,739 records is one gene with a boolean call, 619 of them essential.	3,739	1	3,739	is__essential	scalar	node	1.9×10^{6}	best	1 gene, its own label
GeneEssentialityGoodall2018	*E. coli*	Goodall 2018 built a mini-Tn5 transposon library of about 3.7 million mutants in *E. coli* BW25113, sequenced the transposon junctions by TraDIS, and called each protein-coding gene from its insertion index. This store holds the essential and non-essential calls of two conditions, the library as plated and after outgrowth in LB broth, one record per gene and condition.	8,309	2	8,309	is__essential	scalar	node	5.1×10^{6}	best	1 gene, its own label x condition
GeneEssentialityPrice2018Ecoli	*E. coli*	Price 2018 likely-essential genes of the *E. coli* BW25113 RB-TnSeq library KEIO ML9, from Supplementary Table 1: a gene is called essential or nearly essential when the fitness analysis gave it no value and its insertion and read densities were both under 0.2, on LB plates at 37 C. One record is one called gene, 324 in all.	324	1	324	is__essential	scalar	node	2.6×10^{5}	best	1 gene, its own label
GradedGeneEssentialityGoodall2018	*E. coli*	Goodall 2018 TraDIS essentiality of a mini-Tn5 library of about 3.7 million mutants in *E. coli* BW25113, called from each gene's insertion index by a mixture likelihood ratio, as plated on LB agar (TL) and after outgrowth in LB broth. This class holds the 317 calls of unclear, one record per gene and condition, with insertion index and likelihood ratio.	317	2	317	essentiality__call	scalar	node	2.2×10^{5}	best	1 gene, its own label x condition
GradedGeneEssentialityRoyet2025	*P. putida*	Royet 2025 Table S4: the TRANSIT HMM essentiality call of every KT2440 gene from the mariner Tn-seq library, 5,729 genes over two stages, 11,458 records. One record is one gene at one stage, the library as plated on LB agar or after outgrowth in LB, storing the released state with its TA sites, insertion rate and mean read count.	5,729	2	11,458	essentiality__call	scalar	node	4.8×10^{6}	best	1 gene, its own label x condition
Total (100 datasets, 100 probed)					**7,257,468**				7.7×10^{9}		